Founder Memo: Reduce COD RTO Without Killing Growth

To: Founder **Purpose:** Pilot recommendation based on the supplied order sample

Recommendation

Use a four-action risk policy rather than blanket COD bans. Ship low-risk orders; confirm moderate-risk orders; offer prepaid for elevated risk; and hold/cancel only for very high predicted risk combined with prior RTO history, with manual review. Protect first-time customers from automatic cancellation based solely on a model score. Thresholds are pilot settings, not proven optimal cutoffs.

Three findings from the data

Finding	Observed result	Operational implication
Repeat RTO history	34 labeled orders with ≥2 prior RTOs; 76.5% RTO	Use as a verification signal, not an automatic ban.
First-time buyers	1,517 orders with zero prior orders; 30.3% RTO	Most are not RTO; use confirmation/prepaid choices rather than blanket cancellation.
Geographic variation	Bihar: 53.4% across 146 orders; Maharashtra: 17.9% across 390	Investigate carrier/coverage mix; do not block an entire state.
Data quality	30 duplicate order IDs, 12 missing pincodes, 4 missing cities; 1 IN_TRANSIT row excluded	Resolve data quality before production claims.

Policy and latest-month retrospective simulation

Initial thresholds: below 20% risk → ship as-is; 20% to below 36% → WhatsApp confirmation; 36% to below 50% → prepaid nudge; 50%+ plus at least one prior RTO → hold/cancel for review. Thresholds are configurable. In September 2026, the out-of-fold historical simulation assigned 136 orders to ship as-is, 107 to confirmation, 85 to prepaid nudge, and 6 to hold/cancel (334 labeled orders). These are retrospective model assignments—not actual interventions or measured savings.

Money: why a ■30 prepaid offer is a reasonable test

Estimated direct cost per RTO = ■65 forward shipping + ■65 return shipping + ■25 packaging/handling + (8% × ■180 expected damage) = ■169.40. With assumed RTO rates of 28% for COD and 4% for prepaid, the direct RTO-cost reduction per successful prepaid conversion is (28% − 4%) × ■169.40 = ■40.66; after a ■30 discount, this narrow cost-only estimate is ■10.66. It excludes conversion changes, discounts paid to customers who would have delivered anyway, and contribution margin. A delivered ■649 order at 62% gross margin contributes about ■402.38, so cancelling a genuine order can destroy more value than the direct RTO cost avoided. Measure net contribution in a randomized holdout; do not call modeled savings realized profit.

Role of the local LLM

The prototype includes an optional local Ollama model (Qwen2.5 3B) to suggest clarification questions for ambiguous English/Hindi/Hinglish address text. Manual smoke tests found useful flags for “Near station” and a Hinglish landmark address, no ambiguity for a more detailed address, and a misleading response to an inconsistent city/state input. This tiny test set is qualitative only. The LLM is advisory: it cannot change risk scoring, thresholds, pincode validation, financial calculations, or cancellation decisions. It is not a delivery verification service.

Risks and next two weeks

Key risks: duplicate/mislabelled data, model drift, noisy geographic proxies, false address flags, low response/prepaid conversion, discount abuse, and false cancellations. In the next two weeks: (1) reconcile duplicate IDs and outcome definitions; (2) evaluate on a later time period and check calibration; (3) run a randomized holdout with human review of high-risk cases; (4) track RTO, confirmation response, prepaid conversion, cancellation reversals, complaints, and net contribution by customer type, state, courier, and acquisition source; (5) only then scope approved WhatsApp templates, secure payment confirmation, audit logging, and rollback controls.

All financial results are assumption-based estimates. The supplied data is observational and may be synthetic or sampled; causal impact must be measured through a controlled pilot.